

Ten Features of a Data-Generating Process for Simulating Aggregated N-of-1 Trials: A Framework for Comparison

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Purpose. A Monte Carlo study of an N-of-1 design can only show what its data-generating process (DGP) lets the data contain. This paper identifies ten features of a DGP that determine what a simulation of an aggregated N-of-1 trial can and cannot reveal. They can be used to describe, compare and choose among DGPs.

For each feature we give the intuition, the mathematics, the relevant literature, the choices made in this compendium and the evidence for them, and a best-practice recommendation. Section 12 compares five DGPs on all ten features:

- the published process of Hendrickson et al. (2020; “RH”);
- the package’s current construct;
- the refined process proposed in paper 01;
- mean moderation;
- the pharmacodynamic sensitivity model of docs/38.

Section 13 is a summary written for the introduction of paper 01. Evidence labels follow the compendium convention: derived, verified (computed or simulated), inspected, argued.

Notation index and glossary

Notation follows analysis/report/NOTATION.md. Symbols marked *local* are defined here.

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t ; $t = 0$ is baseline	canonical
BL_i	baseline level	canonical
$BR_{it}, PB_{it}, TV_{it}$	drug, expectancy (placebo) and natural-course responses	canonical
B_i, b_i	biomarker; standardized biomarker	canonical
t_{od}, t_{sd}	time on drug in the current on-drug period; time since discontinuation	canonical
$t_{1/2}$	carryover (response) half-life	canonical

Symbol	Meaning	Status
e_t	expectancy: 1 open label, 0.5 blinded, 0 at baseline	canonical
$f(t)$	population drug-response time course, scaled to 1 at full effect	local (docs/38)
g_t	coupling profile: fraction of the biomarker's moderation present at visit t	local (docs/37)
c_{bm}	covariance-moderation strength (correlation)	canonical
β_{bm}	mean-moderation strength (multiplier)	canonical
$\omega_i, \bar{\omega}, \zeta, u_i$	participant drug sensitivity, its mean, the biomarker's effect on it, unexplained sensitivity	local (docs/38, renamed in docs/44)
ρ	AR(1) serial correlation per week	canonical
K, A	component correlation; temporal correlation kernel	local (docs/35)
$\beta_{bm:D}$	biomarker-by-treatment interaction, the estimand	canonical

Glossary.

- **DGP.** The probability model from which simulated trials are drawn.
- **Predictive / prognostic biomarker.** Associated with the size of the treatment effect, or with the outcome irrespective of treatment.
- **Coupling.** How strongly the biomarker is tied to the drug response at a given visit.
- **Ceiling.** The largest biomarker effect for which a covariance-based DGP has a valid joint distribution.

1. Summary

#	Feature	Question it answers	Recommended choice
1	Outcome decomposition	What is the score made of?	Baseline minus drug, expectancy and natural-course responses, each with its own clock
2	Onset of the drug response	How does benefit build on drug?	A saturating time course in time on drug, with clinically read landmarks; an effect-compartment or turnover model as the principled form

#	Feature	Question it answers	Recommended choice
3	Offset and carryover	How does benefit decay off drug?	Pharmacodynamic half-life, anchored at discontinuation, zero before first exposure
4	Re-exposure	What happens when the drug restarts?	Drug-specific: restart from zero for re-titrated drugs, from the residual otherwise
5	Encoding of the predictive biomarker	How is the interaction placed in the model?	A moderation profile that fades with the drug effect, in the mean or the covariance; never a step
6	Prognostic pathways	Can the biomarker act without the drug?	Optional couplings to natural course, placebo response and baseline
7	Within-person dependence	How do deviations persist over visits and across components?	Separable AR(1) plus a person-level term, valid for every schedule
8	Between-person heterogeneity of response	Do participants differ in response beyond the biomarker?	Explicit unexplained sensitivity; drug-response variance that shrinks off drug
9	Observation process	How are the latent responses observed?	Phase-specific measurement error, the visit schedule as data, specified missingness mechanisms
10	Validity, estimand and calibration	Is the DGP coherent, and what does the simulation estimate?	Valid by construction (stop, never repair); an explicit estimand; parameters calibrated and reported

Three cross-cutting findings motivate the list:

1. **Defects in features 3 and 5 together produced the published power collapse under carryover** (docs/37).
2. **Feature 7 bounds what feature 5 can represent** (docs/34 to docs/36).
3. **Feature 6 decides which analysis is valid** (docs/48, docs/49).

A DGP comparison that varies one feature while holding the others fixed is the only way to attribute a result to a modeling choice.

2. Framing: what a DGP must do

A simulation study defines aims, data-generating mechanisms, estimands, methods and performance measures (the ADEMP structure of Morris, White and

Crowther 2019 [1]; see also Burton et al. 2006 [2]). For N-of-1 designs, the DGP must do three things.

- **Represent the clinical phenomena the design is meant to resolve:** within-person contrasts over time, carryover, expectancy, and between-person differences in response.
- **Encode a well-defined estimand,** here the biomarker-by-treatment interaction $\beta_{bm:D}$.
- **Be valid and verifiable.** It must be a proper probability model, with every parameter interpretable and every result reproducible.

Published simulation studies of N-of-1 designs make these choices implicitly and differently [31, 32, 33, 34], and the CENT reporting statement asks for several of them to be reported for real trials [35]. The ten features below are the places where DGPs for these designs actually differ. We selected them by three criteria:

- the choice changes at least one simulation result materially (verified in this compendium);
- it has a pharmacological or clinical interpretation;
- it can be stated and compared across DGPs.

3. Feature 1: outcome decomposition

What it is. The symptom score is built from separately generated components, each with its own driver:

$$Y_{it} = BL_i - (TV_{it} + PB_{it} + BR_{it}).$$

Component	Driver
TV (natural course)	time in the trial
PB (expectancy)	time in the trial, scaled by the expectancy e_t
BR (drug response)	time on drug

Intuition. An N-of-1 contrast compares a participant on drug with the same participant off drug. What makes that contrast informative, or misleading, are the processes that change over the same weeks: the condition improves on its own, belief in treatment produces benefit, and blinding changes that belief. A DGP that generates only “drug effect plus noise” cannot show how a design copes with these.

Mathematics. Each component is observed only through the sum. Identification therefore comes from design variation:

- expectancy varies between the open-label and blinded phases;
- drug state varies within the blinded phase;
- time varies throughout.

In the Hybrid design, the open-label visits are all on drug at $e_t = 1$ and every off-drug visit is blinded at $e_t = 0.5$. Expectancy and drug are therefore partly confounded, and only the blinded visits separate them cleanly ([docs/40](#), Section 7; [docs/49](#), Section 2).

Literature. Decomposing response into drug action and disease progression is standard in pharmacometrics [3, 4], and placebo response has its own longitudinal models [28]. That expectancy adds benefit when a treatment is known to be given is shown by the open-versus-hidden paradigm [29]. Its dependence on trial design (open label against placebo-controlled) is modeled by [30]. The three-component construction for N-of-1 designs is RH’s [5].

This compendium. All DGPs use RH’s decomposition and Gompertz parameters. The component decomposition paper (06) and [docs/31](#) examine whether the components can be recovered.

Best practice.

- Generate each component on its own clock.
- State which design contrasts identify which component.
- Report component-level parameters on the outcome scale.

4. Feature 2: onset of the drug response

What it is. The time course of benefit after the drug is started.

Intuition. Benefit builds over weeks, through titration and the accumulation of effect. Early on-drug visits therefore show a partial effect, and a design whose on-drug periods are short sees mostly partial effects.

Mathematics. RH use a modified Gompertz curve in time on drug,

$$G(t_{od}; m, d, r) = m \frac{\exp(-de^{-rt_{od}}) - e^{-d}}{1 - e^{-d}},$$

with $m = 10.99$, $d = 5$ and $r = 0.42$ per week. Its landmarks:

- the inflection is at $t = \ln d/r = 3.8$ weeks;
- half of the maximum is reached at 4.7 weeks;
- 90% is reached at 9.2 weeks ([docs/40](#), Section 4).

The pharmacometric alternative is a turnover (effect-compartment) model,

$$\frac{df}{dt} = k(E(t) - f(t)), \quad k = \frac{\ln 2}{t_{1/2}^R},$$

in which onset and offset share one rate. On drug from level f_0 , $f(t) = 1 - (1 - f_0) 2^{-t/t_{1/2}^R}$, so onset is exponential rather than sigmoid. A sigmoid onset follows if the target $E(t)$ itself rises with titration.

Literature.

- **Effect-compartment models** describe the lag between concentration and effect [6].
- **Indirect-response (turnover) models** describe responses that build and decay through production and loss [7].
- **E_{max} and sigmoid-E_{max} models** relate exposure to effect [8].

This compendium. RH's Gompertz onset is kept in every current DGP. docs/38 proposes the turnover form, which is not yet implemented.

Best practice.

- **Read the onset clinically** before simulating: how long until half of the effect is reached. Check it against the design's on-drug period lengths.
- **Prefer an onset that is coherent with the offset** (Feature 3).

5. Feature 3: offset and carryover

What it is. How the drug response decays after the drug is stopped.

Intuition. Two kinds of carryover exist, and they act on very different time scales.

- **Pharmacokinetic carryover** is drug still in the body. For prazosin it is about 2.3 hours of plasma half-life [9], gone within a day.
- **Pharmacodynamic carryover** is benefit that persists after the drug has gone, set by how fast the response turns over: days to weeks.

At weekly visits, only the pharmacodynamic kind matters, so the simulation's half-life is a response half-life.

Mathematics. Anchored decay from the level at discontinuation:

$$\mu_{BR,t} = \mu_{BR,\text{stop}} 2^{-t_{sd,t}/t_{1/2}}.$$

RH's recursion instead decays the previous visit's already-decayed mean by the total time since discontinuation. At off-drug visits one and two weeks after stopping, the second is then at $2^{-(1+2)/t_{1/2}}$ instead of $2^{-2/t_{1/2}}$, so the decay compounds (derived; docs/37, Appendix A). Before first exposure $t_{sd} = 0$, but there is no drug effect, so the response is zero. A decay function evaluated at $t_{sd} = 0$ returns 1 and must not be used there.

That error, made in the analysis-side indicator of manuscript 02, cost the exposure-weighted analysis 0.26 to 0.30 in power in the crossover design (verified, docs/45, Section 8).

Literature.

- **Carryover is the classical problem of crossover designs** [10, 11].
- **Behavioral carryover** that outlasts the pharmacological effect has been modeled explicitly [12].
- **Washout and analytical treatments** in N-of-1 designs [13].
- **Causal estimation under carryover** in aggregated N-of-1 trials, with a simulated DGP [27].

This compendium.

- **Decay forms:** exponential, with Weibull as a sensitivity option (manuscript 02, the stress test).
- **Half-life grid:** 0, 0.5 and 1 week.
- **Pharmacological range:** 0.1 to 0.2 week for prazosin (paper 01).

Best practice.

- **Specify carryover as a pharmacodynamic half-life**, with its justification.
- **Anchor the decay at discontinuation.**
- **Code pre-exposure visits as unexposed.**
- **Vary the decay form as a sensitivity analysis.**

6. Feature 4: re-exposure

What it is. The drug response when the drug is restarted after an off-drug period.

Intuition. Either the response rebuilds from zero, which is right if the drug is re-titrated, or it continues from whatever remains of the earlier effect, which is right if the residual carries forward.

Mathematics.

- **Restart from zero:** $\mu_{BR,t} = G(t_{od})$ with t_{od} reset. In the Hybrid crossover, a participant at near-full response at week 10 is credited with $G(4) = 4.28$ of 10.99 at week 16, 39% of the maximum.
- **Turnover model:** restart from the residual is automatic, $f(t) = 1 - (1 - f_{\text{res}}) 2^{-t/t_{1/2}^R}$.
- **Combined rules:** anchored offset with a zero restart can step the mean down at re-exposure, whenever the residual exceeds the restarted curve.

Literature. Re-titration after an interruption is standard prescribing practice for prazosin, because of first-dose hypotension. Turnover models give restart from the residual by construction [7].

This compendium. RH, the package and the refined process all restart from zero. Restart from the residual is roadmap step 7 of docs/44 and is not implemented.

Best practice.

- **Make the rule an explicit, drug-specific choice.**
- **For re-titrated drugs, restart from zero.**
- **Otherwise, restart from the residual,** preferably through a turnover model.

7. Feature 5: encoding of the predictive biomarker

What it is. How the biomarker-by-treatment interaction enters the model. There are three forms:

- **Covariance binding (RH):** a correlation $c_{bm}g_t$ between the biomarker and BR_{it} .
- **Mean binding:** a shift of $\beta_{bm}b_i\sigma_{BR}g_t$ in BR_{it} .
- **Sensitivity binding (docs/38):** $BR_{it} = \omega_i f(t) + \varepsilon_{it}$ with $\omega_i = \bar{\omega}(1 + \zeta b_i) + u_i$.

Intuition. A predictive biomarker tells you how much drug effect to expect. Wherever the drug effect is absent, the biomarker should therefore predict nothing. What matters most is not *where* the interaction is placed but its profile over visits, g_t , and above all whether that profile fades with the drug effect.

Mathematics.

(a) **Conditional equivalence** (derived). Under joint normality,

$$E(BR_{it} | B_i) = \mu_{BR,t} + c_{bm}g_t\sigma_{BR}b_i, \quad \text{Cov}(BR_{it} | B_i) = \Sigma_{BR} - c_{bm}^2\sigma_{BR}^2gg^\top.$$

So covariance binding is mean binding with exposure profile g_t and $\beta_{bm} = c_{bm}$, and a rank-one smaller residual covariance.

(b) Power depends on the profile through the coupling gap (derived, docs/37). For the paired-difference statistic,

$$\beta_{bm:D} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} (1 - \bar{g}_{\text{off}}),$$

where $\bar{g}_{\text{off}}$ is the mean off-drug coupling. Power falls with the gap $1 - \bar{g}_{\text{off}}$, and the required sample size scales roughly as $1/\text{gap}^2$.

Profile	Off-drug coupling	Gap under carryover
RH's step rule	1 wherever any drug effect remains	0: the interaction vanishes at any positive half-life
Graded coupling	$2^{-t_{sd}/t_{1/2}}$	falls smoothly with the half-life
Sensitivity model	$f(t)$, proportional to the response	follows the response

On RH's own code, graded coupling took power from 0.13 back to 0.64 at a half-life of 0.1 week (verified, docs/37, Appendix A).

(c) The ceiling (derived; docs/34, docs/35). Covariance binding requires

$$c_{bm}^2 u^\top M^{-1} u < 1,$$

by the Schur complement, where M is the response correlation block and u the coupling pattern. Under RH's compound symmetry the ceiling is 0.256, below RH's effect sizes of 0.3 and 0.6. RH's code repaired the infeasible matrices silently: 20 of 54 at those effect sizes. Mean and sensitivity binding have no ceiling: under sensitivity binding the correlation is an output,

$$\text{Cor}(B, BR_t) = \frac{\bar{\omega} \zeta f(t)}{\sqrt{(\bar{\omega}^2 \zeta^2 + \sigma_u^2) f(t)^2 + \sigma_\varepsilon^2}}.$$

Literature.

- **Mean moderation** is the standard representation of treatment-effect heterogeneity by a covariate [14, 15].
- **Random participant-by-treatment variation** is the standard description of individual response in replicated crossovers [16, 17].

- **RH were, to our knowledge, the first** to parameterize the interaction as a covariance in the N-of-1 literature [5].
- **Report 03** shows that covariance binding matches the second moments of a latent responder-class mixture, with $c_{bm}^2 = f_B f_{BR}$, the product of the between-class variance fractions.

This compendium.

- **The refined process** uses graded coupling.
- **Paper 01** compares the three encodings.
- **The sensitivity model** is proposed and not yet implemented.

Best practice.

- **Choose the profile first, and make it fade with the drug effect.**
- **Choose the placement second.** Covariance binding is acceptable below its ceiling. Use mean or sensitivity binding when larger effects are needed.
- **Report the strength on two scales:** as the correlation or multiplier, and as the implied slope in outcome points per SD of biomarker.

8. Feature 6: prognostic pathways

What it is. Whether the biomarker can be associated with the outcome through routes that do not involve the drug.

Intuition. Blood pressure might predict who improves on placebo, or who improves anyway. A DGP in which the biomarker touches only the drug response cannot reveal analyses that mistake such an association for a drug interaction.

Mathematics. Each pathway implies a slope of Y on b at baseline (s_0) and afterwards (s_t):

Pathway	s_0	s_t
natural course, constant	0	$-c_{bm,TV}\sigma_{TV}$
natural course, growing	0	$-\beta_{bm}^{TV}\sigma_{TV}G_{TV}(t)/\max G_{TV}$
placebo response	0	$-c_{bm,PB}10e_t$
baseline severity	$c_{bm,BL}\sigma_{BL}$	$c_{bm,BL}\sigma_{BL}$

An analysis that shares one biomarker slope between baseline and the off-drug visits fits $\hat{\beta}_B \approx f s_{\text{prog}}$, for some $0 < f < 1$ set by the weight of the baseline row. It then reports a spurious interaction of about $(1 - f) s_{\text{prog}}$ (heuristic, docs/48, Section 2).

At a prognostic coupling of 0.3, the null rejection rate was 0.10 for a random-intercept-plus-AR(1) analysis and 0.42 for an unstructured covariance. Adding the expectancy terms restored it to 0.040 to 0.047 (verified, [docs/49](#)).

Under separability the joint validity condition generalizes to $c_{bl}^2 + \sum_{cc'} [K^{-1}]_{cc'} v_c^\top A^{-1} v_{c'} < 1$ (derived and verified, [docs/35](#), Section 4.8).

Literature.

- **The predictive/prognostic distinction** is central to the PATH statement [14, 15].
- **Treating baseline as a response** imposes a constancy constraint [18].

This compendium. Prognostic couplings to the natural course and to the placebo response exist in the stress-test driver. A baseline-severity coupling is being run. Only the placebo coupling is in the package (`c.bm.pb`).

Best practice. Include prognostic pathways as standard stress scenarios whenever analyses are compared. A DGP without them can confirm an analysis but never refute it.

9. Feature 7: within-person dependence and cross-component structure

What it is. How a participant’s deviations persist over visits, and how the components co-vary.

Intuition. The structure can carry permanent traits (compound symmetry), fading memory (AR(1)), or both. Separability means one clock for all three components: the link between components fades at the same rate as each component’s own memory ([docs/35](#), “In brief”).

Mathematics.

- **Separable form.** $M = K \otimes A$, with K the component correlation and $A_{ts} = \rho^{|w_t - w_s|}$. Its eigenvalues are products of those of K and A , so it is valid for every visit schedule.
- **The non-separable alternative** fails whenever $\lambda_{\min}(A) \leq (c_1 - c_\times)/(1 - c_\times)$, which happens on dense schedules.
- **RH’s construct** is a sum of a person-level and an occasion-level separable term, each valid.
- **The ceiling factors.** Its schedule term is a sum of per-switch costs, for example $1/(1 - \phi^2)$ for off to on, with $\phi = \rho^{\text{gap}}$.
- **Implied outcome covariance.** With the person-level baseline,

$$\text{Cov}(Y_t, Y_s) = \sigma_{BL}^2 + (s^\top K s) \rho^{|t-s|},$$

a random intercept plus AR(1). That is the structure assumed by the random-intercept-plus-corCAR1 analysis.

Consequences (verified).

- **Power.** At the same c_{bm} , AR(1) data give much less power than compound symmetry: 0.335 against 0.721 in Hybrid at 0.25.
- **Analysis validity.** A compound-symmetry working model applied to AR(1) data rejected a true null 0.15 to 0.16 of the time in the crossover design (docs/45).

Literature.

- **Random effects plus serial correlation plus measurement error** as the general longitudinal model [19, 20].
- **AR(1) with unequally spaced visits** [21].
- **Kronecker (separable) covariances** for multivariate repeated measures [22].

This compendium.

- **RH:** compound symmetry, $\rho = 0.8$.
- **The package:** non-separable AR(1).
- **The refined process:** separable AR(1) at $\rho = 0.7$.

Best practice.

- **Specify the structure generatively** (person-level term plus serial process), so that validity holds by construction.
- **State ρ and its basis.**
- **Compare effect sizes across structures on the slope scale.**
- **Check the analysis model against the structure the DGP implies.**

10. Feature 8: between-person heterogeneity of response

What it is. How much participants differ in their drug response beyond what the biomarker explains, and how the drug-response variance behaves over time.

Intuition. Two participants with the same blood pressure need not respond equally. Under RH's construct, the drug-response component has its full SD of 8 even off drug, where its mean is zero. That is as if participants had a random drug response with no drug present, which adds noise to every on-off contrast.

Mathematics. Under the sensitivity model,

$$\text{Var}(BR_t) = (\bar{\omega}^2 \zeta^2 + \sigma_u^2) f(t)^2 + \sigma_\varepsilon^2,$$

which shrinks to the serial-noise variance as the drug effect washes out. The participant-by-treatment variance σ_u^2 is the quantity that replicated designs estimate [16, 17]. It bounds the attainable biomarker correlation by $\bar{\omega}\zeta/\sqrt{\bar{\omega}^2\zeta^2 + \sigma_u^2}$ (derived, docs/38, Section 5). A latent-class (bimodal) response is a different heterogeneity, with nonlinear conditional mean (report 03).

Literature. Individual response and its variance components in N-of-1 and replicated crossover designs [16, 17, 23].

This compendium. RH, the package and the refined process all use a constant drug-response variance. The sensitivity model makes it proportional to the response. Report 03 studies latent classes.

Best practice.

- **Separate explained from unexplained heterogeneity**, that is, ζ from σ_u .
- **Let the drug-response variance follow the drug effect.**
- **Treat the distribution of response as a stated assumption**, normal or mixture.

11. Feature 9: the observation process

What it is. How the latent responses become recorded data: measurement error, its dependence on the study phase, the visit schedule, the scale, and missingness.

Intuition.

- **Ratings can be noisier in some phases.** Open-label and blinded phases can differ in rating noise, as well as in mean.
- **Visits are not equally spaced.** For example, the Hybrid design measures at weeks 4, 8, 9, 10, 11, 12, 16 and 20.
- **Participants drop out**, sometimes because they are doing badly.

Mathematics.

- **Phase-specific measurement error** adds a variance $\sigma_{\varepsilon, \text{phase}}^2$ to each visit, which a homoscedastic working model omits.
- **Missingness mechanisms.** Under MCAR or MAR, a likelihood analysis with all observed data is valid; under MNAR it is not [24, 25].
- **A MAR hazard** depending on the previous score, $\text{logit } h_t = a + \gamma z(Y_{t-1})$, can be calibrated to a target dropout rate per simulated trial (docs/46).

Literature. Missing-data theory [24, 25]. Informative dropout in N-of-1 designs is the subject of paper 09.

This compendium. RH’s censoring patterns; the stress test’s phase measurement error and its MCAR and MAR dropout; paper 09’s informative dropout. Under 20% MCAR or MAR dropout, the random-intercept-plus-AR(1) analyses with CR2 standard errors held their size (at most 0.068). The published analysis rejected 0.079 under MCAR dropout and 0.071 with phase-specific measurement error. The unstructured model with AIC selection rejected 0.075 under MAR (verified, stress test).

Best practice.

- **Treat the schedule as data.** Use calendar time in every correlation.
- **Include phase-specific error and at least MCAR and MAR dropout** as stress scenarios.
- **State the mechanism of any missingness simulated.**

12. Feature 10: validity, estimand and calibration

What it is. The properties that make a DGP trustworthy as the truth against which analyses are judged.

Intuition. A simulation tests analyses against a stated truth. If the truth silently changes (repaired matrices), is undefined (no estimand), or is unrealistic (uncalibrated parameters), the results cannot be interpreted.

Mathematics and practice.

- **Validity.** Either the DGP is generative (valid by construction), or every matrix is checked and infeasible cells stop the run. RH’s silent repair changed correlations by up to 0.10 at $c_{bm} = 0.6$ (verified, docs/36).
- **Estimand.** State the target, here the on-drug moderation slope, and whether an analysis with a binary drug indicator estimates it or an attenuated contrast.
- **Calibration.** Choose parameters from data where possible, and report effect sizes on the clinical scale. For prazosin, published trials give the plasma half-life [9] and a biomarker association on the Clinician-Administered PTSD Scale [26]. The simulated scale differs, so the mapping must be explicit.
- **Verifiability:**
 - a faithful reference arm reproducing the published process;
 - one change at a time;
 - common random numbers across arms;
 - a Monte Carlo standard error for every reported rate;
 - pre-registration of comparisons [1].

This compendium.

- **Reference arm:** a reproduction of RH's code at commit 58b32a9.
- **Paired designs** throughout.
- **Pre-registered stress test:** docs/46, with one amendment disclosed.
- **Pending:** calibration of effect sizes to trial data.

Best practice.

- **Stop, never repair.**
- **Define the estimand before the DGP.**
- **Keep a reference arm.**
- **Report ADEMP** [1].

13. The five DGPs compared

Feature	RH (58b32a9)	Package covar	Refined (paper 01)	Mean moderation	Sensitivity model (docs/38)
1 Decomposition	BL, TV, PB, BR	same	same	same	same
2 Onset	Gompertz in t_{od}	Gompertz	Gompertz	Gompertz	turnover $f(t)$
3 Offset	compounding recursion	compounding	anchored	anchored or compounding	anchored (turnover)
4 Re-exposure	restart at 0	restart at 0	restart at 0 (option: residual)	restart at 0	from residual
5 Biomarker encoding	covariance, step profile	covariance, graded	covariance, graded	mean, on-drug only (or decayed)	mean through sensitivity, proportional to f
6 Prognostic pathways	none	placebo only	natural course, placebo, baseline (driver)	possible	possible
7 Dependence	CS, $\rho = 0.8$	non-separable AR(1)	separable AR(1), $\rho = 0.7$	as chosen	rank-one plus serial

Feature	RH (58b32a9)	Package covar	Refined (paper 01)	Mean modera- tion	Sensitivity model (docs/38)
8 Response hetero- geneity	constant variance, full off drug	same	same	same	shrinks with the response
9 Obser- vation	censoring sets	censoring sets	phase error, MCAR/MAR (driver)	as chosen	as chosen
10 Validity	silent repair	silent repair in sampler	stop on infeasible	always valid	valid by construc- tion

Driver: implemented in the simulation drivers, not yet in the package.

14. Summary for the introduction of paper 01

Monte Carlo studies of aggregated N-of-1 trials can only show what their data-generating process lets the data contain. We identify ten features on which such processes differ:

1. the decomposition of the score into baseline, natural course, expectancy and drug responses;
2. the onset of the drug response;
3. its offset and carryover;
4. the response on re-exposure;
5. the encoding of the predictive biomarker;
6. prognostic pathways for that biomarker;
7. the within-person dependence over visits and across components;
8. between-person heterogeneity of response;
9. the observation process, including measurement error and missingness;
10. the validity, estimand and calibration of the process.

Three findings motivate the list. First, the power collapse under small carryover reported by Hendrickson et al. is produced jointly by Features 3 and 5. Their biomarker coupling stays at full strength wherever any drug effect remains, so the simulated interaction vanishes at any positive half-life. A coupling that fades with the drug effect removes the collapse. Second, Feature 7 bounds Feature 5. Under compound symmetry the covariance encoding cannot represent biomarker correlations above 0.256, and the published effect sizes relied on silently repaired matrices; a separable AR(1) structure is valid for every schedule and raises the bound. Third, Feature 6 decides which analysis is valid. If the biomarker is also

prognostic, analyses that share its effect with the baseline visit report spurious interactions. We use the ten features to specify a refined process and to compare it with the published one.

15. Limitations

- **The features are drawn from this compendium’s experience** with one disease area (PTSD) and one drug (prazosin). Other settings may add features, for example binary or count outcomes, or multiple biomarkers.
- **Some recommendations rest on derivations** and have not been simulated: the turnover onset, restart from the residual, and the sensitivity model.
- **Parameter calibration to trial data is outstanding.**

16. References

Each reference was verified against PubMed (PMID given) or the publisher’s page, on 2026-10-07. Exceptions are noted.

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